Thomas Henry Devine, B.S., M.S.
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Skills

- Causal Inference (RCTs, CUPED, Synthetic Controls, Difference-in-Differences), experimental design, forecasting, predictive modeling, Monte Carlo, quantitative analysis, data visualization, segmentation, recommender systems, query optimization, reporting infra (CRUD. ETL)

Programming:

- Python (Pandas, Numpy, fastai, prophet, pytorch), R, MySQL, PostgreSQL, Vertica, Snowflake, Hadoop, bash, Git, AWS (S3, Airflow, ECS, RDBS), Postman, Superset, Tableau, Looker, Metabase, Hubspot, Salesforce, MLflow, MLOps, Grafana, Google Workspace

Work Experience

Senior Data Scientist at InsightVectors **Jun 2022-Present**

I delivered data-driven solutions to diverse industries bridging strategies across stakeholder projects, contributing to their success.

- For **Meta**, I worked with the Product Marketing Management team and XFNs (marketing, media, etc) to develop and launch the 2025 H1 campaigns for Brazil Privacy and UK Privacy ($1M) & Personalization ($725k); the goal is to increase user awareness of privacy customization and personalization benefits on-platform in priority markets by targeting low-confidence, skeptical users
- For **Meta**, I collaborated with XFNs to get inferred models (iPersonalization p15-p55, iPrivacy p15-p65) to estimate audience segments; I deduplicated segments across two UK campaigns to preserve causality. With colleagues, I developed the Tessa configs and prepared the Brand Lift Study & Simon Survey questions and translations for BR; I calculated the campaign budget dependent on audience size, frequency, MDE, normalized cost (per investment efficiency framework), and more; I worked with Media to ensure Measurement Plans reflected campaign goals and requirements for budget requests
- Analyzed various High Income News Engaged (HINE) audience definitions for the 2025-H1 USA campaign ($40M+).
- For meta, I worked on a Product-Market Fit analysis for Threads. The project had three parts: (1) what tool should we use for (2) clustering popular content with (3) reporting on topic-subtopics for the most engaged users on Threads; working on using simclusters algorithm by X to model author-to-author content clusters with audience demographic breakdowns (age, gender)
- Engineered an XGBoost model for a job board's Sales team that classified early signs of churn for 19.5k business accounts saving $33.7M in lost SaaS subscriptions revenue through early intervention. Results were served in CSMs' dashboards who would call customers and discuss how to increase their business' utilization of the learning service
- Managed an A/B testing platform for a 7-figure user base for a top 3 sports game client, boosting daily assignments by 12% (134k). I conducted 143 A/B tests, achieving ARPU lifts of 0.4–2.4% and reducing player churn by 3.2%
- Ran a MCMC simulation for an add-on insurance product adding $600,000 in revenue; product later developed per my findings
- Collaborated with leadership to build data products for operations, marketing, and analytics, enhancing productivity, reducing ticket turnaround, and increasing visibility on KPIs and product health and utilization benchmarks
- Maintain & develop bespoke ETL & CRUD pipelines for business-critical automated reporting & migrations. Service reads/writes from/to MySQL (OLTP) & Vertica (OLAP); with batch processing using alembic, yaml, Python (pandas, serde, sqlalchemy), and data in JSON format. I extended features to have logging, reads/writes to AWS S3, email notifications, & more
- Developed a Markov Chain Attribution Model with HubSpot CRM data, improving conversions and lead monetization / revenue downstream by ~2.1%

Senior Data Scientist at Torch Logistics **Apr 2021-May 2022**

As the first Data Scientist, I worked directly with the C-suite to understand the business problems to develop the data KPIs and data foundations for this logistics 3pl. From Q1-2021 to Q2-2022, run rate grew over 703%, carrier network grew over 1020%, and customer network grew 410%

- Designed and deployed rule-based recommendation engine for carrier discovery utilizing multiple public and private data; increased cold-start success rate from 3% to 97% with Sales team feedback loop from functionality in G-Forms, daily crons & AirFlow DAGs in Python with postgreSQL as the OLAP. Results surfaced in Metabase to sales rep-specific dashboards
- Developed a boosted trees pricing model in Python; improved the MAPE from 25% to 14%. The goal was to develop a pricing model that could provide competitive yet profitable rate suggestions for shippers, balancing limited data and high variability
- Built AirFlow & S3 based ETL pipelines to ingest public data (MCMIS) for models and dashboards for the operations team
- Built a business-critical request for pricing (RFP) bidding, revenue analysis, and visualization process in Python

Analytics Consultant at THD Analytics **Aug 2018-Apr 2021**

I advised e-commerce SMBs on a project basis. I helped them make data-driven decisions by processing & modeling their data

- Solved ad hoc requests for: customer preference profiling, customer journey analysis, sales forecasting, and demand planning
- Developed business intelligence product offering (KPI dashboard suite) and regression pricing scheme based on past and current market trends using sales data using SQL & Python. Made a self-service platform for our stakeholders to act on
- Built benchmarking dashboards using Metabase (SQL) to manage workloads and metrics; recommended new metrics for KPIs

Education

M.S. in Economics at Carnegie Mellon University
Awards: $138,000 National Science Foundation GRFP in Economics for a recommender system in child adoption
B.S. in Mathematics at University of North Dakota
B.A. in Economics at University of North Dakota
Awards: Ronald E. McNair Scholar, selected speaker at UND's Phi Beta Kappa induction ceremony